

Linking Funding Mechanisms in Off-Chain Perpetual Futures to the DeFi Leverage Cycle

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Abstract

Little work has been done to potentially explain relations between funding rates of perpetual futures in the centralized exchange (CEX) and the broader DeFi leverage cycle. We construct a new funding rate-derived metric called “Sign-Aware Time to Zero Funding” (SATZeF) as a candidate for an off-chain regime indicator of on-chain leverage, proxied by liquidation volumes. Through empirical observations of lending activity on Aave v3 Mainnet and off-chain data for ETH and BTC on Kraken and Coinbase (CEXs), we show that the SATZeF serves as a reliable off-chain regime indicator of the DeFi ecosystem’s current state with respect to the leverage cycle, while daily funding rates can better explain granular movements in on-chain liquidation volumes. We also show that near-zero correlation between spot prices and funding rates implies that severe liquidations in the short-term are highly unlikely. Using the empirically proven conclusions, we construct a macro-policy framework for monitoring the DeFi leverage cycle from off-chain and on-chain perspectives. Such a framework paves the way for greater insight into potentially enforcing leverage requirements on CEXs as a means of indirectly affecting funding conditions in DeFi lending pools.

Keywords: leverage cycle; decentralized finance; perpetual futures; funding rate; lending pools

JEL Classification: G19; E32; G12

1 Introduction

Decentralized Finance (DeFi), directly opposed to trusted and centralized Traditional Finance (TradFi), refers to a multitude of trustless network avenues providing lending/borrowing pools and decentralized exchanges (DEXs) with automated market makers known as “liquidity pools.” These decentralized lending pools allow users to borrow cryptocurrency tokens, usually *stablecoins* (which are pegged to the dollar), against collateral and income-seeking lenders to deposit tokens that provide liquidity to the lending pool.

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Interestingly, Saengchote (2023) notes that high-end loan value concentration is higher for decentralized lending pools than in TradFi bank systems. However, like all credit markets in TradFi, lending pools are incredibly susceptible to the dangerous effects of liquidity crunches, systemic contagion, and the leverage cycle.

Open borrow positions in lending pools can be *liquidated* by automated liquidators, who are given an incentive by the pool’s protocol Perez et al. (2021). To demonstrate the susceptibility of lending pool liquidity to changes in collateral asset prices, Perez et al. (2021) finds that variations of only 3% of an asset’s price can result in \$10 million in positions being liquidatable. Liquidators are given an economic incentive by giving them an opportunity to buy a portion of the liquidatable collateral at a discount, to subsequently sell it in the open market at a higher price. If the difference between the effective price at the open market and the discounted price exceeds transaction fees, the liquidator has every incentive to perform the liquidation (Tian and Zhu, 2025). Qin et al. (2021) shows, across major lending protocols (Aave, Compound, MakerDAO, and dYdX), that if many borrow positions present a case where transaction fees outweigh the liquidation incentive, the protocol racks up *Type II* bad debt, which can result in dangerous insolvency issues for the pool. Other research efforts have also shed light into the various components of the leverage cycle and systemic risk in lending pools. Saengchote (2023) empirically validates the claim that during periods of high volatility, lender-side demand for stablecoin deposits increases and during periods of high volatility, borrower-side demand for stablecoin loans decreases. These findings from various efforts collectively align with the leverage theory in TradFi.

Though decentralized liquidity and lending pools are subjects of great interest for research, *centralized exchanges* (CEXs) for token trading comprise a large portion of the cryptocurrency market. Furthermore, though distinct, DEXs and CEXs are very much linked and share propagating effects through arbitrage effects. To demonstrate this, Aquilina et al. (2023) analyzes the propagating effects of liquidations; they find that deleveraging in DeFi markets corresponds to lower prices and that on-chain deleveraging laggedly propagates to off-chain exchanges. Additionally, Chiu et al. (2025) shows that the pledgeable value of collateral in lending pools and the asset’s own value in centralized exchanges are determined together, showing that cross-market activity tightly link both DEXs and CEXs.

Furthermore, a large portion of CEX venues consist of *perpetual futures* markets on various tokens. Due to the nature of funding mechanisms and high leverage capacity of perpetual futures, they represent instruments of speculation and indicative of the centralized leverage cycle. Perpetual futures venues in CEXs incorporate *autodeleveraging* (ADL), a last-resort socialization mechanism implemented by the CEXs when insurance funds are insufficient to restore solvency to the venue; they work by haircutting profits from top-performing accounts through a queue-based algorithm (Chitra et al., 2026). Chitra (2025) was the first to provide a rigorous ADL model, finding that moral hazard grows asymptotically as venue participation scales. Additionally, Inan (2025) analyzed the predictability of funding rates in perpetual futures markets through a *double autoregressive model* (DAR); a key research question (RQ) left unanswered, but acknowledged by the author, within this paper was that there was notable correlation between BTC/USDT spot prices and the corresponding perpetual futures funding rate series in various episodes; their modeling choices also showed that modeling stablecoin price series, explored by Djogbenou et al. (2023), share striking similarities with funding rates (with the most no-

table being that both exerting mean-reverting behavior). Similarly, Werapun et al. (2025) explores the risk and return profiles of funding rate arbitrage, finding that BTC demonstrated significant correlation between spot price movements and funding rate returns; however, they also note that for the XRP coin, funding rates and spot prices are almost completely unrelated.

The seminal TradFi paper by Geanakoplos (2009) provides a unified theory of the leverage cycle in centralized markets, with notable contributions such as (a) the policy implication to curtail leverage in ebullient economic periods and to prop up leverage in anxious times and (b) the introduction of *debt overhang*, the phenomenon where investors (such as homeowners and banks) no longer have a good incentive to invest or make loans even if they haven’t defaulted prior. The work of Geanakoplos builds off of Kiyotaki and Moore (1997), who were among the first to show that a fall in collateral prices results in a fall in overall borrowing capacity, assuming a constant *loan-to-value* (LTV) ratio. Additionally, Aymanns and Farmer (2015) built an agent-based model to show that pro-cyclical leverage behavior in institutional agents (when perceived risk is low, leverage is high and vice versa), the leverage cycle (i.e. gradual increases in stock prices and leverage are followed by severe market collapses and deleveraging) endogenously arises from such behavior; they also find that countercyclical regulation policies can significantly reduce the amplitude of the leverage cycle. Similarly, Nuño and Thomas (2017) also studies the endogenous leverage cycle, proposing general equilibrium framework and find that leverage constraints are endogenously imposed on banks in the presence of risk-shifting moral hazard.

Ultimately, a multitude of research efforts have studied the leverage cycle in the traditional and centralized banking system. On the other hand, DeFi research have seen limited insight into the leverage cycle specifically. Previous efforts have described the mechanics of liquidations and systemic contagion in decentralized lending pools (Aquilina et al., 2023; Perez et al., 2021; Qin et al., 2021). However, to the best of our knowledge, barely any of these efforts link liquidations and risk contagion to a broader procyclical leverage pattern clearly present in the DeFi ecosystem, just as in TradFi markets. Little work has been done with respect to connecting decentralized pool dynamics to centralized perpetual futures venues. The funding rate mechanisms of these centralized perpetual futures yield significant information about the cryptocurrency market’s position with respect to the leverage cycle.

We propose an econometric relation between the perpetual futures avenue and DeFi lending pools. Specifically, we construct a new variable we call “Sign-Aware Time to Zero Funding” (SATZeF). We conduct spectral analysis with a rigorous methodology to avoid spectral leakage and noise on funding rate dta to conclude the existence of cyclicity in funding rates, which justifies its candidate status in serving as a leverage cycle indicator. We compare the performances of daily funding rate data and daily SATZeF as off-chain indicators of the contemporary regime with respect to the DeFi leverage cycle, proxied by liquidation volumes. We subsequently regress both SATZeF and funding rates, using time lags and target transformations, individually against liquidation volumes to compare the performances of both feature variables as an “indicator for granular movements” in the market’s position with respect to the leverage cycle.

Ultimately, we find that the constructed SATZeF metric is a significantly reliable off-chain regime indicator of the DeFi leverage cycle and outperforms daily funding rates in this regard, while daily funding rates are shown to better explain granular movements in daily liquidation volumes than are SATZeF values. These results are conclusive proof

that funding rate mechanisms in off-chain perpetual futures venues within CEXs are tightly coupled with leverage patterns in on-chain lending pools. We also show that near-zero correlation between spot prices and funding rates implies that severe liquidations in the short-term are highly unlikely. Using the empirically proven conclusions, we finally construct a macro-policy framework for monitoring the DeFi leverage cycle from off-chain and on-chain perspectives. Such a framework paves the way for greater insight into potentially enforcing leverage requirements on CEXs as a means of indirectly affecting funding conditions in DeFi lending pools.

This paper is outlined as follows. Section 2 describes the econometric model linking perpetual futures in CEXs and DeFi lending pools, Section 3 describes data collection and exploratory data analysis of the data collected on funding rates and DeFi liquidation volumes, Section 4 discusses the methods used to evaluate the econometric model, Section 5 shows the results of the empirical testing of the econometric model, and Section 6 ends the paper with a discussion of the obtained results with tangible international macro-policy implications.

2 Model

Let f_i denote the funding rate, corresponding to a chosen liquid CEX and tokenized asset, at time t_i for $i = 0, 1, 2, \dots$ to reflect the discrete observational nature of funding rates in perpetual futures markets. Assume that f_i is an hourly observation series, which is consistent with the data collected and detailed in Section 3.1; that is, $f_0, \dots, f_{24}$ encompasses the entire funding rate history within the first day. We introduce a new variable coined “Time to Zero Funding” (TZeF). At time t_i , the TZeF is a function of the funding rate time-series up to time t_i , meaning it is adaptive to the information space up to the present time. Let τ_i denote the TZeF at time t_i . It’s formula is given by Equation (1).

$$\tau_i = \begin{cases} 0 & \text{if } a(i) \\ i - \max_{\substack{j < i \\ a(j)}} & \text{otherwise} \end{cases} \quad (1)$$

where $a(i)$ is a condition on i and f_i given by

$$a(i) = \min_{i - (i \bmod 24) \leq i \leq i - (i \bmod 24) + 24} [f_i] < \epsilon \quad (2)$$

where ϵ is some chosen small constant. Semantically, Equation (1) translates to the TZeF being the number of days since the last day during which the hourly funding rate was very close to zero (the threshold for “zeroness” defined by ϵ_{DF}) on at least one timestep. The final variable used in our econometric model is a sign-transformed version of the TZeF coined “Sign-Aware Time to Zero Funding” (SATZeF), denoted by $\theta(\tau_i) = \theta_i$. Specifically, if the arithmetic mean of hourly funding rates, ranging from the most recent time k when $\tau_k = 0$ to the current time i , is negative, then $\theta_i = -\tau_i$. Otherwise, $\theta_i = \tau_i$. This sign transform encodes the direction of funding rates into θ , while maintaining the historical information implicitly encoded into the original TZeF variable τ .

We also construct the “spot-funding correlation” variable, which we define as the rolling-window (with a chosen window of 7 days) Pearson correlation between spot prices and funding rates. As Inan (2025) was among the first to find notable contemporary correlation

between spot prices and funding rates, we anticipate that it will have significant relation with the DeFi leverage cycle.

The notion that centralized venues are linked with decentralized liquidity and lending pools is an established one (Cong et al., 2025; Aquilina et al., 2023; Chiu et al., 2025), so the extension to funding rates being an indicator of not just the centralized leverage cycle but the leverage cycle of the entire DeFi ecosystem is not an unprecedented one. We note that such arbitrage and leverage bounded relations between centralized and decentralized venues may largely arise from price oracles within decentralized lending pools, which use USD price quotes of tokens in centralized exchanges such as Coinbase to determine value of collateral in the lending pools.

Our proposed econometric model consists of two parts: (1) a *regime indicator* (RI) of the broad position of the DeFi ecosystem with respect to the leverage cycle, and (2) an off-chain *indicator for granular movements* (IGM) in ecosystem-wide leverage. We compare SATZeF and daily funding rate data for an RI of the broader DeFi leverage cycle that extends to on-chain activity away from centralized venues determining these funding rates in the first place. Specifically, let daily USD liquidation volumes, denoted by V_t , be the direct on-chain indicator of the DeFi leverage cycle. Then, suppose we create deterministic two competing definitions of leverage cycle regimes:

1. At time t_i , $\theta_i > 0$ indicates abundant leverage in the DeFi ecosystem, $\theta_i = 0$ indicates a slight absence of leverage and possible a transitional step between the leverage cycle regimes, and $\theta_i < 0$ implies a dangerous amount of deleveraging and liquidations and an absence of leverage in the DeFi ecosystem.
2. At time t_i , $f_i > 0$ indicates abundant leverage in the DeFi ecosystem, $f \approx 0$ indicates a slight absence of leverage and possible a transitional step between the leverage cycle regimes, and $f < 0$ implies a dangerous amount of deleveraging and liquidations and an absence of leverage in the DeFi ecosystem.

Meanwhile, we also compare SATZeF and funding rates for a reliable and accurate IGM for leverage in the DeFi ecosystem. Specifically, we resolve to determine which of the two would better explain movements of daily liquidation volumes than the other. Through empirical observations, we effectively compare SATZeF and funding rates for serving as reliable, off-chain indicators of the DeFi leverage cycle.

Note that we do not form a definition of leverage cycle regimes for spot-funding correlation, as there is little research that has been done with respect to this variable that would allow for an informed a priori hypothesis about its relation with respect to the DeFi leverage cycle.

3 Data

3.1 Data Collection

For daily on-chain liquidation volume on ETH, we use the “Aave v3 Mainnet” subgraph from The Graph, using GraphQL.¹ Note that Aave v3 Mainnet resides in the Ethereum

¹This subgraph is publicly invokable using an API key at <https://thegraph.com/explorer/subgraphs/Cd2gEDVeqnjBn1hSeqFMitw8Q1iiyV9FYUZkLNRcL87g?view=Query&chain=arbitrum-one>

Time-series	Test Statistic	p -value
Funding rate	-12.680	0.000
Liquidation volume	-27.093	0.000
Spot-funding correlation	-30.757	0.000
SATZeF	-6.009	0.000

Table 1: Results of the Phillips-Perron Tests

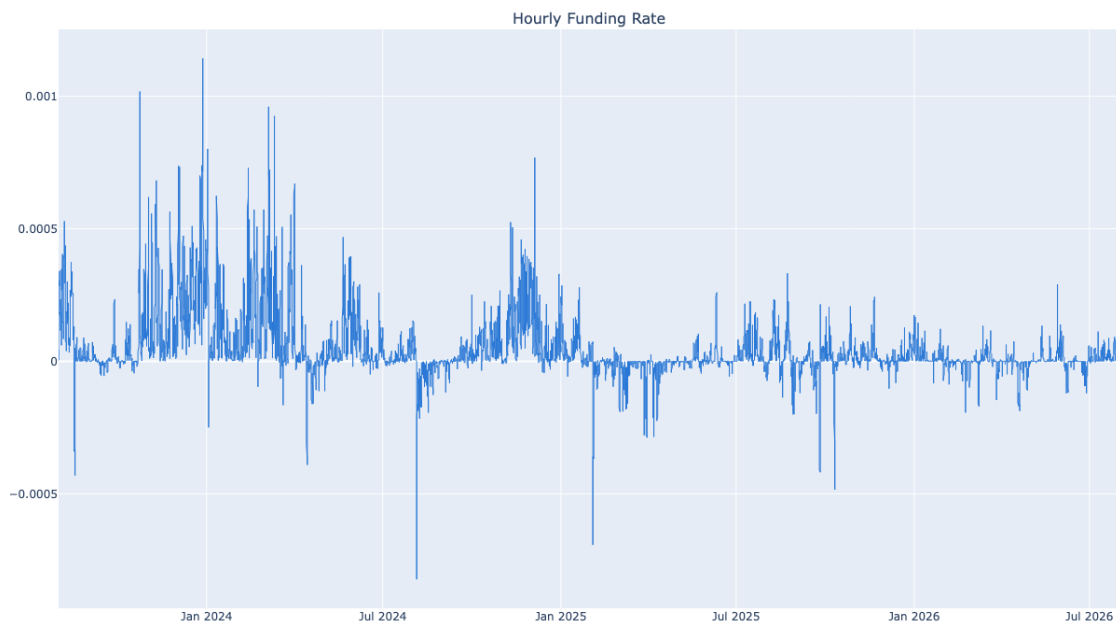
(ETH) blockchain and is subject to lending transactions with a variety of tokens, including ETH, Wrapped Bitcoin (WBTC), and stablecoins. Note that WBTC is an Ethereum token with a 1:1 peg on the native Bitcoin (BTC) price. We use a free and publicly available REST API from Coinbase for daily ETH and BTC spot prices. We use a REST API provided by Kraken, which uses a funding interval of 1 hour, for retrieving raw funding rates corresponding to ETH and BTC perpetual futures. Our sample ranges from 2023-08-01 to 2026-07-31. Additionally, using the definition of SATZeF in Section 2, we construct a daily SATZeF series within the chosen date horizon using $\epsilon_{DF} = 5 \times 10^{-6}$ for ETH and $\epsilon_{DF} = 1 \times 10^{-6}$ for BTC, which we empirically observe to be a reasonable estimate of “zerness” in the funding rate time-series for the two tokens. A time-series graph of the SATZeF is shown in Figure 2. Using obtained funding rates and spot prices, we construct the spot-funding correlation time-series defined in Section 2 as well.

3.2 Exploratory Data Analysis

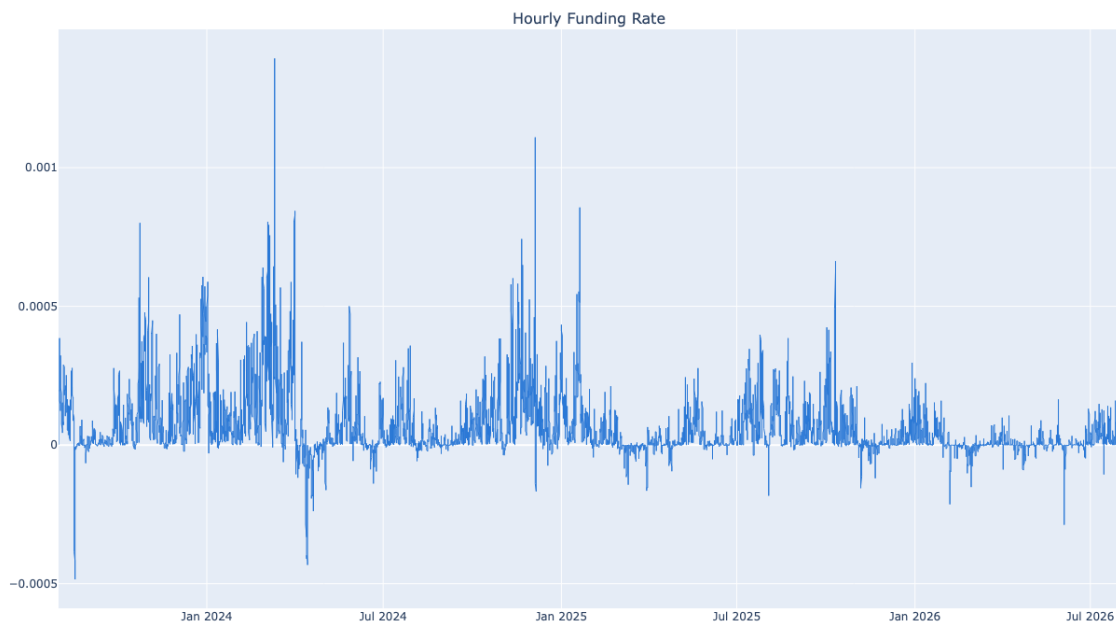
We first conduct exploratory data and spectral analysis on $\{f_{t_i}\}_i$ and $\{\theta_{t_i}\}_i$ to gain insight into the possibly cyclical and diffusive nature of these time-series data. Aligned with anecdotal evidence, we visually observe from Figure 1 that funding rates (for both tokens) are positive most of the time, with the exception of occasional downward, negative spikes. Note that by virtue of the speculative nature of perpetual futures, a consistently positive funding rate implies a greater amount of leverage that traders undertake within the centralized venue.

We first conduct the Phillips-Perron test, which accounts for weak intra-dependence or autocorrelation, on both $\{f_{t_i}\}_i$ and $\{\theta_{t_i}\}_i$ individually to test for weak stationarity (i.e. constant statistical properties of first and second order moments); in the Phillips-Perron test, the null hypothesis is that the time-series process contains a unit root, and the alternative hypothesis is that the time-series is weakly stationary (Phillips and Perron, 1988). From Table 1 that shows the results of these Phillips-Perron tests, we can statistically conclude weak stationarity for all of the time-series processes that we are working with.

We now conduct spectral analysis on funding rates to prove its periodicity justifying the procyclicality of its transformation into SATZeF. We compute the sample periodogram graph of demeaned hourly funding rates for estimates of its power spectral density (PSD) at various frequencies. We use the Welch method coupled with Hanning windows to minimize noise and spectral leakage (Jwo et al., 2021). Figure 4 shows the graph of the computed periodogram of the observed demeaned hourly funding rates for both ETH and BTC perpetual futures. Fascinatingly, the Fourier frequency with the highest PSD, observable by the peak in the periodogram, is computed to be 0.00391 for *both* ETH and BTC funding rates. The ETH dominant peak has a PSD value of approximately

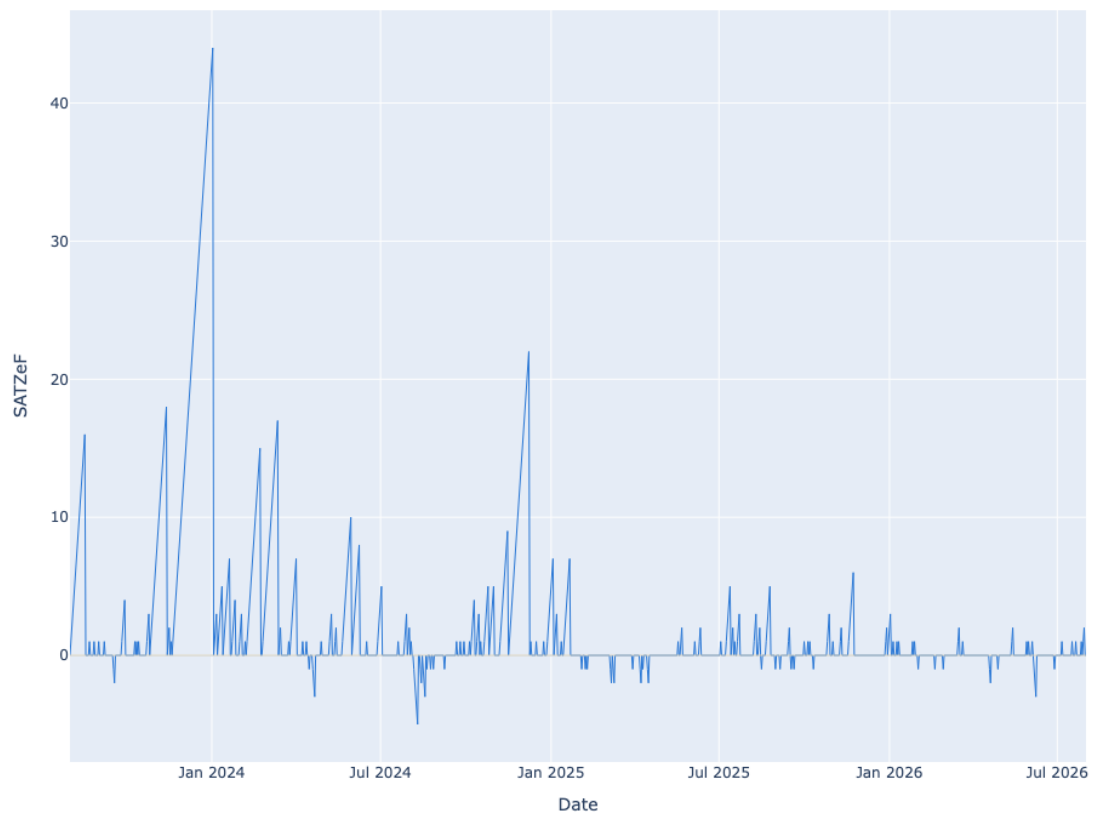


(a) ETH

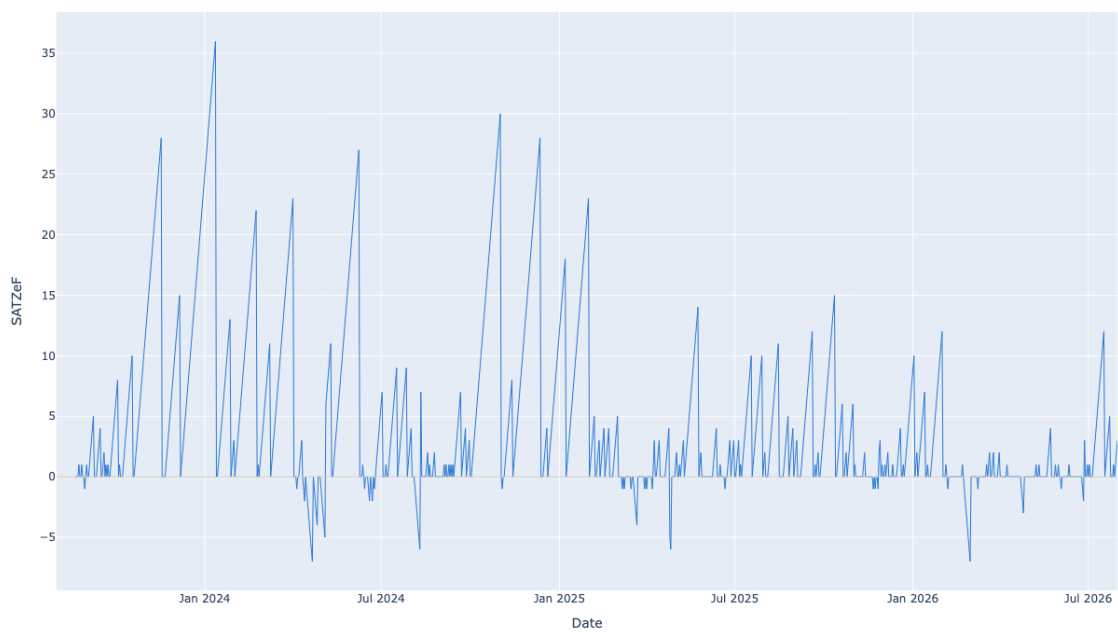


(b) BTC

Figure 1: Hourly Funding Rates



(a) ETH



(b) BTC

Figure 2: SATZeF

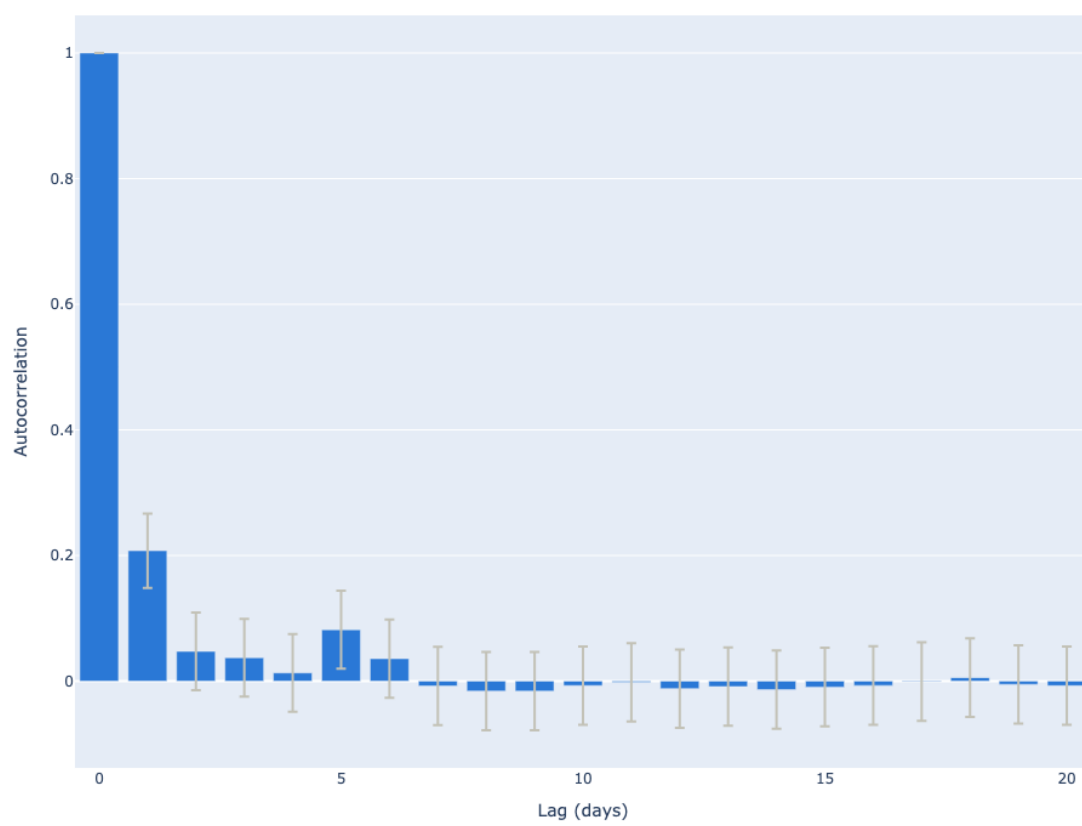


Figure 3: Sample ACFs of Liquidation Volumes (USD)

2.94×10^{-7} , while that of the BTC dominant peak is approximately 3.20×10^{-7} . This dominant frequency corresponds to the dominant cyclical period of 256.00 hours or 10.67 days. To determine whether there is significance in this peak, we use Fisher’s g-test (Quinn, 2021). Fisher’s statistic is calculated as the peak-to-total ratio, given by

$$g = \frac{\max_{1 \leq j \leq m} I(F_j)}{\sum_{j=1}^m I(F_j)} \quad (3)$$

where n is the sample size, $F_j = j/n$ is the j -th Fourier frequency, $I(F_j)$ is the periodogram evaluated at F_j , and $m = \lfloor (n-1)/2 \rfloor$. The p -value can be subsequently calculated as

$$p = \sum_{j=1}^{\lfloor 1/g \rfloor} (-1)^{j-1} \binom{m}{j} (1-jg)^{m-1} \quad (4)$$

Using Equations (3) and (4), we calculate $g \approx 0.1838$ and $p\text{-value} = 8.091 \times 10^{-10}$ for ETH funding rates, and $g \approx 0.2020$ and $p\text{-value} = 4.603 \times 10^{-11}$ for BTC funding rates. Thus, we conclude the existence of a statistically significant dominant sinusoidal frequency in the hourly funding rate time-series in both tokens.

We comment on the results of the Fisher’s g-test. Though the hourly funding rate time-series (see Figure 1) of both tokens can occasionally diverge (especially during tail events), both funding rate series exhibit the same exact dominant harmonic frequency (10.67 days); since ETH and BTC are perhaps the most liquid and widely used tokens in the DeFi ecosystem, this dominant frequency is likely applicable to most liquid tokens for which perpetual futures are available within solvent, centralized venues.

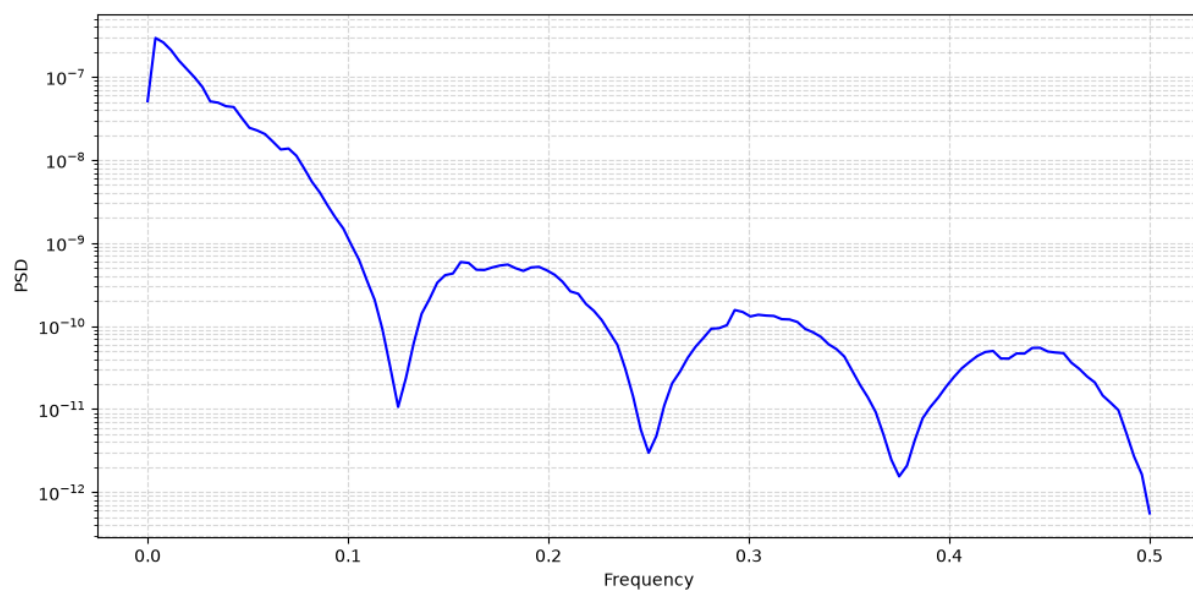
4 Methods

4.1 Comparing RI Indicators

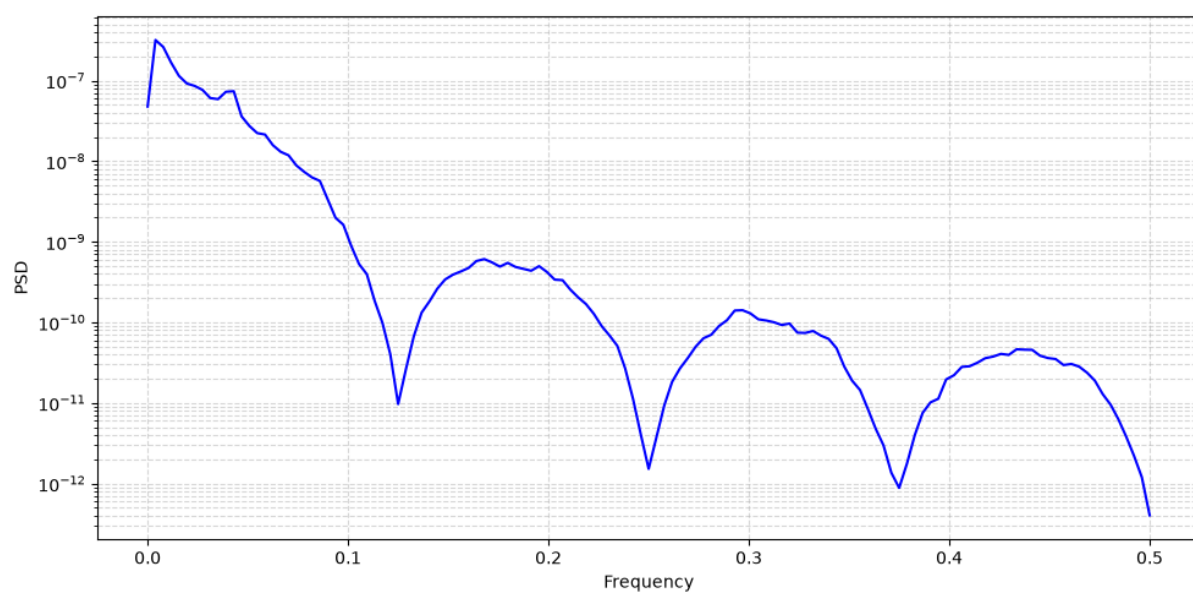
To compress hourly funding rate data to a daily time-series, we take the minimum funding rate within each day. We do this to capture the tail behavior of funding rates, appropriate for macro-analysis of leverage cycles in DeFi, conducive to tail events such as liquidation cascades. However, we note that through sample statistics, significant difference is not found between using the arithmetic mean of the hourly funding rates within a day, or using the minimum or maximum.

With the daily funding rate DF_t as the RI, the macro-leverage regimes are split into three phases defined as follows:

1. $DF_t > \epsilon_{DF} \implies$ abundant leverage prevails in the DeFi ecosystem.
2. $|DF_t| < \epsilon_{DF} \implies$ a slight absence of leverage exists, along with the possible of a transitional step between the two primary leverage cycle regimes.
3. $DF_t < -\epsilon_{DF} \implies$ the DeFi ecosystem is subject to a dangerous amount of deleveraging and liquidations, along with largely an absence of leverage in the DeFi ecosystem.



(a) ETH



(b) BTC

Figure 4: Periodograms of Demeaned Hourly Funding Rates

In our empirical tests, we maintain the same $\epsilon_{DF} = 5 \times 10^{-6}$ used to construct the SATZeF time-series.

On the other hand, with the daily SATZeF as the RI, the macro-leverage regimes are split into three phases defined as follows:

1. $\text{SATZeF}_t > 0 \implies$ abundant leverage prevails in the DeFi ecosystem.
2. $\text{SATZeF}_t = 0 \implies$ a slight absence of leverage exists, along with the possible of a transitional step between the two primary leverage cycle regimes.
3. $\text{SATZeF}_t < 0 \implies$ the DeFi ecosystem is subject to a dangerous amount of deleveraging and liquidations, along with largely an absence of leverage in the DeFi ecosystem.

For spot-funding correlation as the RI, we observe the following partitions:

1. $(\text{spot-funding correlation})_t > \epsilon_{SF}$
2. $|(\text{spot-funding correlation})_t| < \epsilon_{SF}$
3. $(\text{spot-funding correlation})_t < -\epsilon_{SF}$

Based on visual inspection of spot-funding correlation graphs, we choose $\epsilon_{SF} = 0.01$ for a fair estimate of “zerness” for spot-funding correlations. For all three RI-candidates, we compute the sample means of liquidation volume under each regime for a preliminary comparison between the RI performances of the two. To find statistical significance in the differences between these sample means (thus proving the efficacy of the RI indicator candidate), we implement block bootstrapping due to the presence of covariance between data within regimes and covariance between regimes themselves, thus creating a messy covariance structure around which a seamless block bootstrapping resampling method can straightforwardly navigate (Carlstein, 1986). We use the *circular bootstrapping* (CB) method, coined by Politis and Romano (1992), in which the distribution of the block size is a unit mass at a chosen block size and zero everywhere else; this contrasts with *stationary bootstrapping*, in which the underlying block distribution is geometric with the mean equal to a chosen block size (Politis and Romano, 1994). Since the liquidation volume time-series is stationary, we are permitted to use block bootstrapping.

Our CB procedure involves choosing an optimal block size, the method of which will be described later, and construct 100,000 resamples of the liquidation volume time-series data across the entire time horizon. Within each resample, for both the funding rate *and* the SATZeF as the RI, we compute (a) the difference between the liquidation volume sample mean within Regime 1 (abundant leverage) and the liquidation volume sample mean within Regime 2 (slight absence of leverage), and (b) the difference between the liquidation volume sample mean within Regime 3 (severe lack of leverage) and the liquidation volume sample mean within Regime 2. Similarly, within each resample, for spot-funding correlation as RI, we compute (a) the difference between the liquidation volume sample mean within Regime 1 (positive spot-funding correlation) and the liquidation volume sample mean within Regime 2 (near-zero spot-funding correlation), and (b) the difference between the liquidation volume sample mean within Regime 3 (negative spot-funding correlation) and the liquidation volume sample mean within Regime 2. Across all three

RI candidates, the sample mean difference in (a) will be referred to as “Positive-Zero Difference”, and the sample mean difference in (b) will be referred to as “Negative-Zero Difference.” We record the distribution of these sample mean differences, compute the proportion of sample mean differences over (or under, depending on the sign of the mean of sample mean distribution) 0. If this proportion is under the chosen significance level of $\alpha = 0.05$, then we conclude a statistically significant difference in liquidation volume means. We denote this proportion by $P(\text{null})$.

The validity of a central limit theorem for the sample mean is given by a weak mixing condition, which essentially means that autocorrelation function decays with increasing time lags, as iterated by Politis and White (2006), select formulas of which were provided corrections for by Patton et al. (2009). This mixing condition is clearly satisfied for the liquidation volume time-series by inspecting Figure 3, where there is only a significant non-zero ACF for time lags of 0 (obviously), 1, and 5, and the ACF can be clearly seen to decay to near-zero. Let $R(k)$ denote the sample ACF of liquidation volumes with time lag k . Accordingly, Politis and White (2006) provides a detailed methodology of computing the optimal CB block size b_{opt} , given by Equation 5.

$$b_{opt} = \left[\left(\frac{2G^2}{D} \right)^{1/3} N^{1/3} \right] \quad (5)$$

where

$$D = 2g^2(0), \quad (6)$$

$$G = \sum_{k=-M}^M \lambda(k/M) |k| R(k), \quad (7)$$

$$g(\omega) = \sum_{k=-M}^M \lambda(k/M) R(k) \cos(\omega k), \text{ and} \quad (8)$$

$$\lambda(k) = \begin{cases} 1 & \text{if } |k| \in [0, 1/2] \\ 2(1 - |k|) & \text{if } |k| \in [1/2, 1] \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Note that $M = 2\hat{m}$, where $\hat{m}$ is the time lag after which the ACF appears negligible. By this definition, we choose $M = 12$ as a conservative estimate by inspection of the sample ACFs of the liquidation volumes (see Figure 3).

4.2 Comparing IGM Indicators

Determining the performance of the two IGM indicator candidates involves regressing, through ordinary least squares (OLS) linear regression, each of them against daily liquidation volumes. For each candidate, we perform this regression in a wide range of time lags added to the feature variable (i.e., a negative time lag means that the feature lags behind the target variable) and report the correlation coefficient r and p -value associated with each time lag. Of course, to account for autocorrelation in the time-series involved, we implement the Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors of Newey and West (1987), provided out-of-the-box by the ‘statsmodel’ library in Python.

We choose a time lag range of $-20, -19, \dots, 0$ added to the feature variable to examine potentially contemporary *and* predictive relationships between the chosen candidates and liquidation volume. For each of the three IGM candidates (SATZeF, funding rate, and spot-funding correlation), we also conduct an additional log-transform of the target variable for exponential regression, to determine if the feature-target relationships may be nonlinear.

5 Results

We note that RI and IGM results for both ETH and BTC off-chain data (spot prices and funding rates) are strikingly similar, leading to and corroborating the same final conclusions equally. We report results for only ETH in this section, and the corresponding results of BTC can be found in Appendix A for RI results and Appendix B for IGM results.

5.1 RI Indicators

The sample means for the three regimes for both the funding rate and SATZeF as RI indicators are given in Table 2. Evidently, the SATZeF as the RI produces much starker differences in liquidation volume sample means among the three regimes than does the funding rate. A visual histogram of the sampling distributions of liquidation volume mean differences for both indicators are given in Figures 5 and 6. The computed $P(\text{null})$ values for both candidate indicators are given in Table 6, where we can see that the SATZeF indicator produces p -values less than 0.05 for both Positive-Zero Differences and Negative-Zero Differences, whereas the funding rate indicator yields a statistically significant p -value only for Positive-Zero Differences. The inferiority of funding rate as RI is further corroborated with BTC results (see Tables 15 and 13), where the funding rate seems to yield lower liquidation volumes in Regime 2 than Regime 1 along with a high (insignificant) p -value. Since the SATZeF as the RI indicator captures more significant differences of liquidation volumes within the two end leverage regimes from liquidation volumes within the intermediate leverage regime, we can safely conclude that the SATZeF is a better RI indicator than is the funding rate.

Additionally, from Figures 5 and 6, we can see that the sampling distributions for both SATZeF and funding rate as the RI have varying spreads and precision. Thus, we compute the interquartile range (IQR) of each distribution and produce it in Table 7. From the produced IQR table, we observe that both SATZeF and funding rate as RIs produce similarly narrow ranges of Positive-Zero Differences. This implies that both RI candidates are precise (but not accurate in the case of funding rate) about their estimates of the difference between average liquidation volumes during abundant leverage and average liquidation volumes during a slight absence of leverage. Meanwhile, the SATZeF produced an IQR for Negative-Zero Differences that is nearly triple that of the funding rate. This implies that though the both the SATZeF and funding rate are satisfactory RIs of deleveraging regimes, the funding rate may hold an advantage over the SATZeF of producing more precise and less variable estimates of the difference between average liquidation volumes during severe deleveraging and average liquidation volumes during a slight absence of leverage. Despite this precision advantage, due to the comprehensive significance coverage of the SATZeF as RI (as evident in Table 6), it remains the superior

Regime	Funding Rate	SATZeF
Abundant leverage (1)	\$180,709.87	\$190,298.19
Slight absence of leverage (2)	\$173,130.33	\$849,004.64
Severe lack of leverage (3)	\$1,944,660.57	\$3,636,190.94

Table 2: Liquidation Volume Sample Means Across the Three Leverage Regimes

Regime	Sample Mean
Positive Spot-Funding Correlation	\$991,171.26
Near-zero Spot-Funding Correlation	\$93,075.27
Negative Spot-Funding Correlation	\$461,451.51

Table 3: Liquidation Volume Sample Means for Spot-Funding Correlation as RI Across its Three Regimes

candidate for a general RI of the leverage cycle.

Additionally, at first glance, the 0% p -values obtained for spot-funding correlations as RI in Table 6 provide promising results for its efficacy as an RI. However, upon inspection of Figure 7 and the sample means with spot-funding correlation as RI in Table 3, we see that *both* the tail regimes, $(\text{spot-funding correlation})_t > \epsilon_{SF}$ and $(\text{spot-funding correlation})_t < -\epsilon_{SF}$, capture heightened liquidation volumes with respect to the intermediary regime $|(\text{spot-funding correlation})_t| < \epsilon_{SF}$. To explore this discrepancy, we compute the 95% confidence interval of mean liquidation volumes (generated by the same CB method), with all three candidate RIs in their corresponding three regimes and produce it in Tables 4 and 5. We can observe that for both funding rate and SATZeF as RIs, the liquidation volume mean of the computed confidence interval increases when moving from Regimes 1 (positive RI value) to 3 (negative RI value). However, in the case of spot-funding correlation as RI, liquidation volumes are fairly high in the positive RI value regime (with a tail mean of over a million and a half USD), drops to staggeringly low liquidation volumes in the near-zero RI value regime (with a right-tail mean of a measly \$6,395.22), and shoots back up to very high liquidation volumes in the negative RI value regime.

Near-zero spot-funding correlation corresponding to a contemporary near-zero liquidation volume as well is a fascinating finding. Of course, increasing the chosen ϵ_{DF} value will certainly increase the average liquidation volumes corresponding to near-zero spot-funding correlation, but the results shown produce a conclusive proof that contemporary deleveraging is least likely when the average premium index of the perpetual futures contract moves independently of the underlying spot price.

5.2 IGM Indicators

We produce the results, with correlation and p -values, of the HAC-adjusted OLS regression of the two IGM indicator candidates (funding rate and SATZeF) against both the regular and log-transformed liquidation volume.

We direct our attention to the IGM results of the funding rate first. Table 8 shows the OLS results with and without the log-transformation of the target variable. We observe

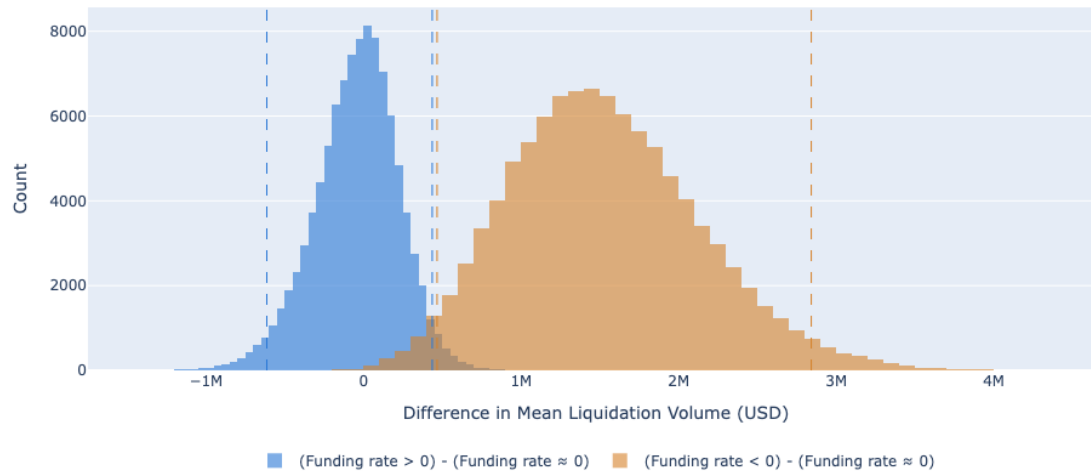


Figure 5: Resampling Distribution of Liquidation Volumes Mean Differences with Funding Rate as RI

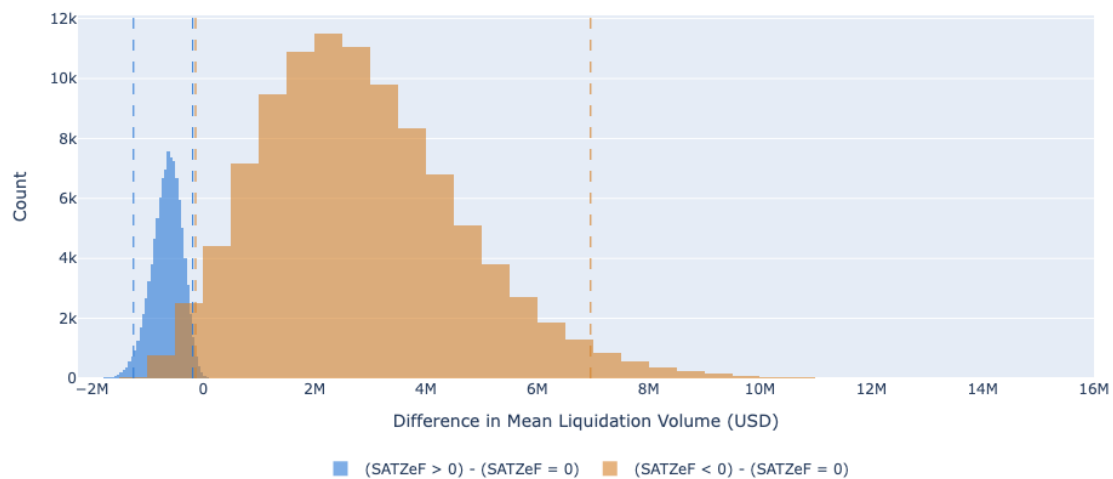


Figure 6: Resampling Distribution of Liquidation Volumes Mean Differences with SATZeF as RI

	Regime 1	Regime 2	Regime 3
Funding Rate	(\$70,010.87, \$376,535.46)	(\$394,453.39, \$1,437,497.27)	(\$672,007.06, \$7,788,836.00)
SATZeF	(\$159,306.70, \$833,460.18)	(\$122,983.14, \$1,056,586.55)	(\$915,424.73, \$3,324,397.53)

Table 4: 95% Confidence Interval Mean Liquidation Volume for Funding Rate and SATZeF as RIs

	Positive Regime	Near-Zero Regime	Negative Regime
Spot-Funding Correlation	(\$459,075.73, \$1,659,339.91)	(\$70.27, \$6,395.22)	(\$209,376.57, \$787,979.05)

Table 5: 95% Confidence Interval Mean Liquidation Volume for Spot-Funding Correlation as RI

that for contemporary movements (0 time lag), the standard, un-transformed OLS is superior (funding rates having a low-moderate correlation with liquidation volumes) over the log-transformed regression for funding rate as IGM. Furthermore, because of the proven periodicity of funding rates, past funding rates are also significantly correlated with contemporary liquidation volumes. Indeed, we observe that the log-transformed OLS (exponential regression) produces correlations of slightly greater magnitude than does the un-transformed OLS for lagged (with time lags of 1-3 days) funding rates (i.e. when funding rate is predictive of liquidation volumes), which is also true in the case of BTC (see Table 17). Thus, these results also yield conclusive proof that funding rates in the past 1-3 days have some significant predictive power over contemporary liquidation volumes.

We now direct our attention to the IGM results of SATZeF. Table 9 shows the OLS results with and without log-transformation of the target variable. Unlike the case with funding rates as IGM, log-transformation of liquidation volumes significantly improves correlation between SATZeF and the target variable for both contemporary *and* lagged co-movements. All time lags with the log-transformed OLS with SATZeF as IGM produce very low p -values, allowing us to conclude significant exponential correlation as is with the case of funding rate as IGM. However, unlike funding rate as IGM, the correlation between SATZeF and log-transformed liquidation volume is fairly invariant to time lag, with correlations staying steady within -9.5% to 14.9%. These conclusions are consistent with BTC results as well (see Table 18).

For funding rate and SATZeF as IGMs, most HAC-adjusted p -values were lower than $\alpha = 0.05$ to conclude significance. However, across un-untransformed and log-transformed and these two IGM candidates, the most meaningful extracted correlation (-36.1976%) was between the non-lagged funding rate and un-transformed liquidation volume. On another note, the lagged correlations had an oscillating tendency and some of them yielded correlations with high enough magnitudes to prove utility in macro-DeFi analysis and policy. Ultimately, we conclude that the funding rate is superior to the SATZeF as a reliable IGM for the DeFi leverage cycle.

Lastly, we direct our attention to the IGM results of spot-funding correlation. Table 10 shows the OLS results with and without log-transformation of the target variable. Unlike with the case of funding rate and SATZeF as RIs, we produce R^2 and the sign of correlation since the HAC-adjusted R^2 is negative in some places. With the target liquidation volumes log-transformed, only the un-lagged feature produces a significant p -value; with

	Funding Rate	SATZeF	Spot-Funding Correlation
Positive-Zero Difference	45.76%	0.131%	0.00%
Negative-Zero Difference	0.065%	3.312%	0.00%

Table 6: RI Results: $P(\text{null})$ Values

	Funding Rate	SATZeF	Spot-Funding Correlation
Positive-Zero Difference	\$340,667.06	\$365,835.21	\$420,221.11
Negative-Zero Difference	\$817,843.09	\$2,405,836.00	\$214,714.65

Table 7: RI Results: IQR Values

the target liquidation volumes un-transformed, the spot-funding correlation produces no significant relation with liquidation volumes anywhere (with any time lag). Thus, though spot-funding correlations produce fascinating results as an RI, it does not possess any meaningful power in explaining granular movements of liquidation volumes. This is corroborated by the fact that a significant correlation was captured nowhere in the BTC results for spot-funding correlation as IGM (see Table 19).

6 Discussion

Throughout the exploratory data analysis of funding rates, RI analysis, and IGM analysis, we have conclusively proven the existence of *cyclical* behavior in funding rates and that off-chain funding rates are deeply aligned with on-chain lending activity.

Recall that CEXs, although somewhat misaligned with the original objectives of decentralized blockchain ledgers, are integral to the DeFi system, particularly through price oracles in DeFi lending pools (Nadler et al., 2026). These relations are likely and partly from (a) implicit associations through leverage undertaking and speculative borrowing and (b) direct relations (through price oracles) between DeFi lending pools and CEX spot prices, which are directly related to funding rates as empirically observed (Inan, 2025). While CEXs, in both spot and perpetual futures avenues, are integral to DeFi activity, they are also establishing themselves as recognized entities under regulatory scrutiny. On May of 2026, the United States Commodity Futures Trading Commission (CFTC) gave approval for a Bitcoin perpetual futures contract to be listed under a CFTC-regulated exchange (Commodity Futures Trading Commission, 2026).

Interestingly, Barbon et al. (2026) empirically demonstrates that DeFi interest rates can sometimes diverge significantly from established policy rates. Liang (2026) shows that stablecoin yields still track behind Fed policy rates, but by a significant $T + 3$ structural lag (universal for both compliant USDC and unregulated USDT stablecoins). Therefore, a clear economic disconnection between DeFi and traditional banking has always existed. However, with perpetual futures (a core part of the DeFi leverage cycle as proved in this paper) becoming an established market in TradFi, it is evident that coordinating maximum leverage requirements across CEXs internationally could significantly curtail leverage during unsustainably abundant leverage regimes and increase leverage during dangerous absences of leverage. And with over \$80T in annual trading volume of cryp-

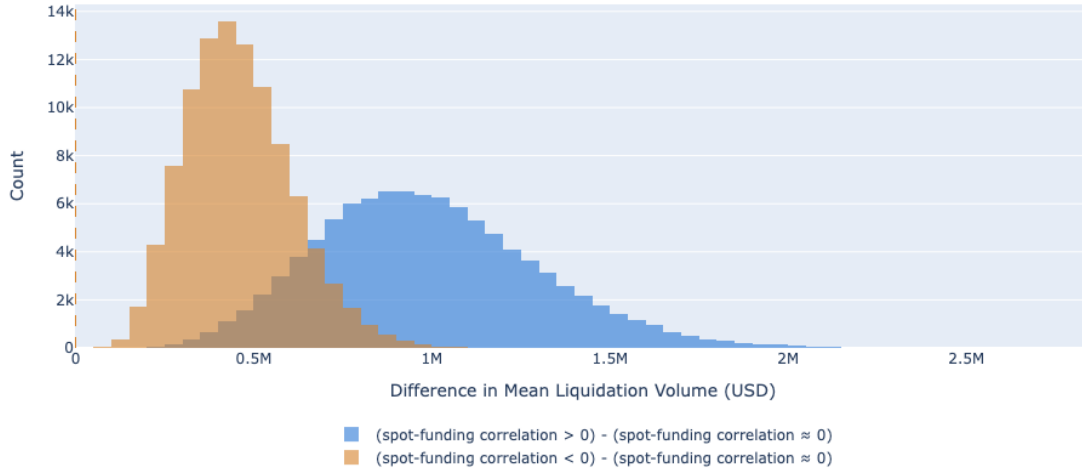


Figure 7: Resampling Distribution of Liquidation Volumes Mean Differences with Spot-Funding Correlation as RI

to currency perpetual futures in 2025 (CoinGecko, 2025), its sheer abundance of presence in the DeFi ecosystem would likely yield effective results for macro-policy coordination. Indeed, the countercyclical policy tool reduces the amplitude of leverage swings in the cycle, as shown by Aymanns and Farmer (2015), and leverage requirements in CEXs can serve as said policy tool to ameliorate the DeFi leverage cycle.

But the primary obstacle preventing such coordinated policy action with perpetual futures CEXs is the absence of an international organization, the jurisdiction of which comprehensively spans all countries with significant DeFi activity, with authority to enforce leverage requirements to these CEXs. Even in the U.S., the Fed rarely updates margin requirements for broker-dealers to curtail the tail effects of the leverage cycle, a key structural flaw in macro-policy design that (Geanakoplos, 2009) has long advocated against. Perhaps in TradFi, an argument could be made that more attention must be paid to short-term funding and money multiplier effects. However, a key incentive for less risk-averse traders to engage in the DeFi ecosystem is the vastly more abundant amount of leverage that is available to the trader by perpetual futures CEXs. Thus, the policy implications for the leverage cycle in DeFi undoubtedly cannot be ignored, due to the speculative nature of the DeFi ecosystem. Therefore, as the DeFi ecosystem grows and becomes a more undeniable portion of the world economy, more attention must be made to the prospect of an international body with proper regulatory authority to enforce these leverage requirements, treating CEXs more like broker-dealers whose margin requirements are subject to the influence of this macro-policy body. By virtue of the three explored metrics serving as an efficient bridge between off-chain perpetual futures trading and on-chain DeFi borrowing-lending activity, such a regulatory body can count on the following macro-policy framework to help monitor and manage the amplitude of the DeFi leverage cycle:

1. $\text{SATZeF} > 0 \implies$ There is an abundant amount of offered and demanded leverage in the DeFi (on-chain and off-chain) ecosystem.

Un-transformed Target			Log-Transformed Target		
Lag (days)	Correlation	p -value	Lag (days)	Correlation	p -value
-20	-0.061888	0.022850	-20	-0.076076	9.519899×10^{-2}
-19	-0.070452	0.009859	-19	-0.098566	2.524992×10^{-2}
-18	-0.061284	0.036313	-18	-0.113394	6.008536×10^{-3}
-17	-0.053241	0.081537	-17	-0.101207	5.097188×10^{-3}
-16	-0.031944	0.136662	-16	-0.141443	7.917038×10^{-5}
-15	-0.026442	0.080513	-15	-0.091865	1.173515×10^{-2}
-14	-0.045000	0.060809	-14	-0.069466	7.281666×10^{-2}
-13	-0.038502	0.036072	-13	-0.083701	1.917457×10^{-2}
-12	-0.061355	0.145805	-12	-0.102818	6.268946×10^{-3}
-11	-0.050052	0.093207	-11	-0.077199	2.021474×10^{-2}
-10	-0.044424	0.046909	-10	-0.104594	1.031714×10^{-2}
-9	-0.039305	0.072847	-9	-0.107868	3.564048×10^{-3}
-8	-0.037268	0.103501	-8	-0.091987	2.139100×10^{-2}
-7	-0.047532	0.014416	-7	-0.099067	1.789936×10^{-3}
-6	-0.069340	0.006686	-6	-0.123923	1.137517×10^{-3}
-5	-0.100609	0.018322	-5	-0.123955	1.167185×10^{-3}
-4	-0.123818	0.008040	-4	-0.144762	9.904921×10^{-5}
-3	-0.130847	0.007767	-3	-0.199645	9.784446×10^{-11}
-2	-0.104262	0.005841	-2	-0.187469	3.239114×10^{-9}
-1	-0.206176	0.000552	-1	-0.210363	2.947478×10^{-12}
0	-0.361976	0.020355	0	-0.216082	5.335071×10^{-11}

Table 8: Correlations and p -values across lagged days for the funding rate as IGM

2. $\text{SATZeF} = 0 \implies$ Potentially lack of leverage. Monitor granular movements in funding rate and liquidation volumes.
3. $\text{SATZeF} < 0 \implies$ Severe lack of leverage with potentially ADLs in perpetual futures venues.
4. Additionally, spot-funding correlation $\approx 0 \implies$ Liquidations or deleveraging (ADLs) are highly unlikely to occur in the short-term monitored horizon.

Thus, by the words of Geanakoplos (2009), “Despite the cries of newspapers to lower the interest rates, the Federal Reserve (Fed) would sometimes do much better to attend to the economy-wide leverage and leave the interest rate alone.” We thus leave it to international regulatory authorities that may form in the future to potentially integrate the proposed macro-policy framework for the DeFi leverage cycle, for policy-wise leverage adjustment likely prove even more pertinent to the DeFi ecosystem than to traditional banking.

Un-transformed Target			Log-Transformed Target		
Lag (days)	Correlation	<i>p</i> -value	Lag (days)	Correlation	<i>p</i> -value
−20	−0.044943	0.003710	−20	−0.142504	2.452268×10^{-6}
−19	−0.048118	0.007985	−19	−0.149088	9.375166×10^{-7}
−18	−0.041511	0.005156	−18	−0.143448	1.358704×10^{-5}
−17	−0.041751	0.004822	−17	−0.138097	2.668693×10^{-5}
−16	−0.043291	0.004123	−16	−0.123683	6.301921×10^{-4}
−15	−0.047142	0.005943	−15	−0.118312	3.135433×10^{-3}
−14	−0.047117	0.004289	−14	−0.107991	1.578690×10^{-2}
−13	−0.054424	0.015798	−13	−0.106128	1.167819×10^{-2}
−12	−0.051622	0.009163	−12	−0.095051	3.787007×10^{-2}
−11	−0.048412	0.005433	−11	−0.104468	1.332809×10^{-2}
−10	−0.049740	0.003854	−10	−0.110960	6.117705×10^{-3}
−9	−0.055891	0.008843	−9	−0.122173	1.739262×10^{-3}
−8	−0.050784	0.004888	−8	−0.119312	1.768628×10^{-3}
−7	−0.048494	0.002763	−7	−0.129213	7.331679×10^{-4}
−6	−0.049859	0.004046	−6	−0.142223	2.373157×10^{-4}
−5	−0.064173	0.025160	−5	−0.140777	3.213532×10^{-4}
−4	−0.064110	0.017586	−4	−0.146206	9.346598×10^{-4}
−3	−0.060501	0.017855	−3	−0.155023	4.478614×10^{-4}
−2	−0.058360	0.016558	−2	−0.135819	1.899145×10^{-3}
−1	−0.056554	0.009832	−1	−0.127158	6.631638×10^{-3}
0	−0.048339	0.005495	0	−0.116623	4.419365×10^{-3}

Table 9: Correlations and *p*-values across lagged days for SATZeF as IGM

Un-transformed Target				Log-Transformed Target			
Lag (days)	R^2	sign(R)	p -value	Lag (days)	R^2	sign(R)	p -value
-20	0.000229	1.0	0.191005	-20	-0.000588	--	0.458196
-19	-0.000910	--	0.894833	-19	-0.000519	--	0.478811
-18	0.000026	1.0	0.293370	-18	0.001490	1.0	0.082013
-17	-0.000163	--	0.287929	-17	0.000640	1.0	0.067204
-16	-0.000086	--	0.294232	-16	-0.000617	--	0.501144
-15	0.000181	1.0	0.183436	-15	0.000447	1.0	0.181370
-14	-0.000818	--	0.721418	-14	0.000171	1.0	0.208476
-13	0.001540	1.0	0.026618	-13	-0.000569	--	0.499356
-12	0.003044	1.0	0.036050	-12	0.002163	1.0	0.081071
-11	-0.000147	--	0.312807	-11	0.000054	1.0	0.278031
-10	-0.000879	--	0.776449	-10	-0.000481	--	0.452682
-9	-0.000247	--	0.332852	-9	-0.000876	--	0.818627
-8	-0.000921	--	0.995217	-8	-0.000435	--	0.443757
-7	-0.000440	--	0.350288	-7	-0.000864	--	0.787199
-6	-0.000919	--	0.973562	-6	-0.000797	--	0.763020
-5	-0.000909	--	0.926081	-5	0.000262	1.0	0.325461
-4	-0.000699	--	0.618687	-4	0.000028	-1.0	0.329510
-3	-0.000916	--	0.973698	-3	0.000570	-1.0	0.233964
-2	-0.000914	--	0.958789	-2	0.000846	-1.0	0.251440
-1	-0.000468	--	0.499484	-1	0.000630	-1.0	0.125479
0	0.001754	1.0	0.074015	0	0.003174	1.0	0.036648

Table 10: R^2 , sign of correlation, and p -values across lagged days for spot-funding correlation as IGM

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A RI Results for BTC

Regime	Funding Rate	SATZeF
Abundant leverage (1)	\$122,712.79	\$190,298.19
Slight absence of leverage (2)	\$274,268.45	\$849,004.64
Severe lack of leverage (3)	\$2,153,106.13	\$3,636,190.94

Table 11: Liquidation Volume Sample Means Across the Three Leverage Regimes

Regime	Sample Mean
Positive Spot-Funding Correlation	\$955,054.52
Near-zero Spot-Funding Correlation	\$20,554.06
Negative Spot-Funding Correlation	\$511,386.83

Table 12: Liquidation Volume Sample Means for Spot-Funding Correlation as RI (BTC) Across its Three Regimes

	Regime 1	Regime 2	Regime 3
Funding Rate	(\$162,118.64, \$675,987.08)	(\$48,883.51, \$508,137.28)	(\$1,105,429.47, \$5,226,662.21)
SATZeF	(\$71,923.51, \$175,066.43)	(\$514,553.87, \$1,767,261.77)	(\$107,286.18, \$21,339,806.46)

Table 13: 95% Confidence Interval Mean Liquidation Volume for Funding Rate and SATZeF as RIs (BTC)

	Positive Regime	Near-Zero Regime	Negative Regime
Spot-Funding Correlation	(\$413,577.22, \$1,629,724.49)	(\$296.35, \$3,241.24)	(\$240,760.39, \$859,152.22)

Table 14: 95% Confidence Interval Mean Liquidation Volume for Spot-Funding Correlation as RI (BTC)

	Funding Rate	SATZeF	Spot-Funding Correlation
Positive-Zero Difference	23.443%	0.213%	0.00%
Negative-Zero Difference	0.008%	2.436%	0.00%

Table 15: RI (BTC) Results: $P(\text{null})$ Values

	Funding Rate	SATZeF	Spot-Funding Correlation
Positive-Zero Difference	\$240,304.74	\$399,693.45	\$420,221.11
Negative-Zero Difference	\$1,361,795.87	\$3,997,798.14	\$214,714.66

Table 16: RI (BTC) Results: IQR Values

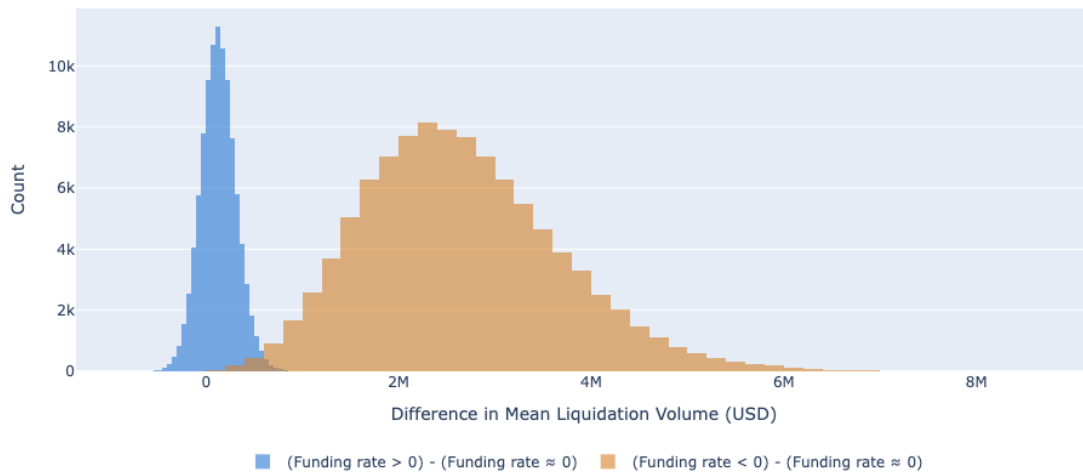


Figure 8: Resampling Distribution of Liquidation Volumes Mean Differences with Funding Rate as RI (BTC)

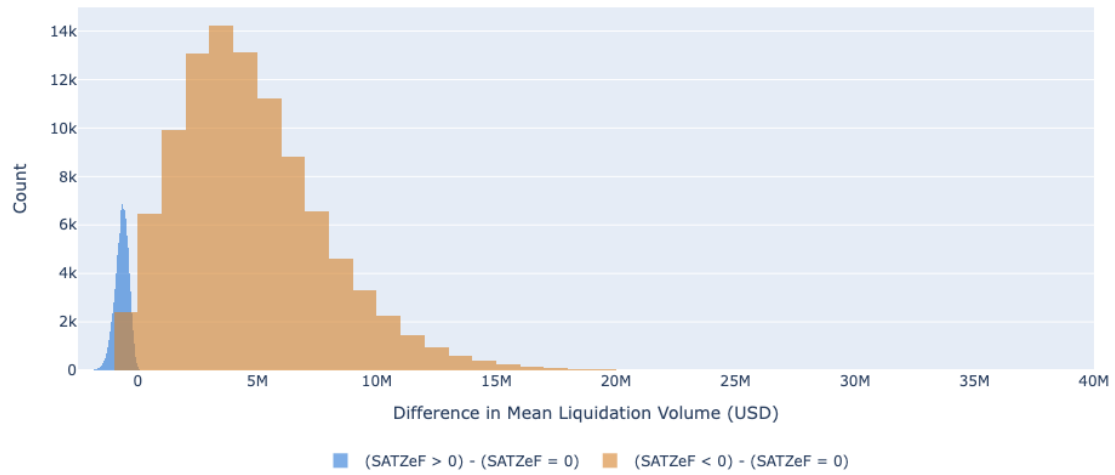


Figure 9: Resampling Distribution of Liquidation Volumes Mean Differences with SATZeF as RI (BTC)

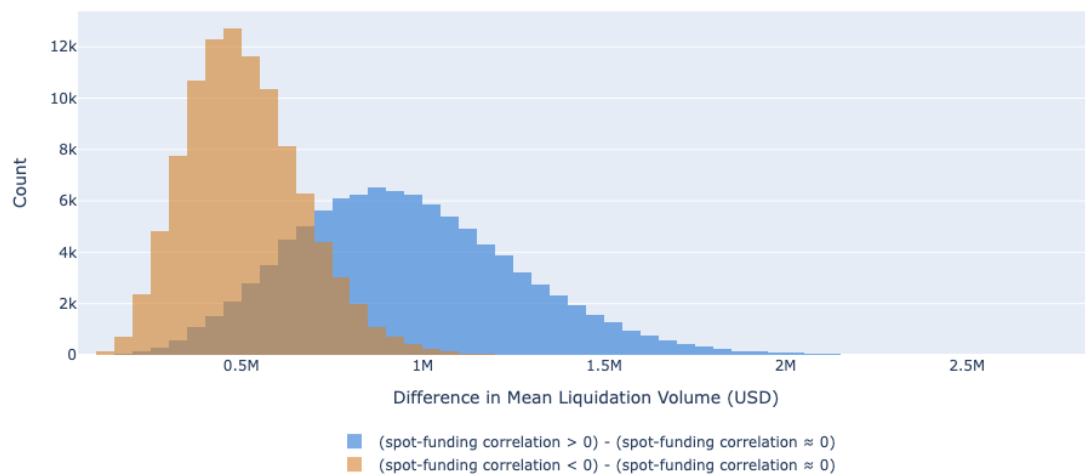


Figure 10: Resampling Distribution of Liquidation Volumes Mean Differences with Spot-Funding Correlation as RI (BTC)

B IGM Results for BTC

Un-transformed Target			Log-Transformed Target		
Lag (days)	Correlation	p -value	Lag (days)	Correlation	p -value
-20	-0.064535	1.444900×10^{-2}	-20	-0.118863	6.525000×10^{-3}
-19	-0.066488	2.192400×10^{-2}	-19	-0.099388	3.073400×10^{-2}
-18	-0.058028	2.567700×10^{-2}	-18	-0.103560	2.561400×10^{-2}
-17	-0.051456	6.261000×10^{-3}	-17	-0.089495	2.878900×10^{-2}
-16	-0.044356	1.370000×10^{-2}	-16	-0.110840	1.342100×10^{-2}
-15	-0.037499	1.940700×10^{-2}	-15	-0.068276	1.296880×10^{-1}
-14	-0.046432	1.580000×10^{-2}	-14	-0.063646	1.151820×10^{-1}
-13	-0.032729	9.983700×10^{-2}	-13	-0.043126	1.657270×10^{-1}
-12	-0.026227	3.993400×10^{-2}	-12	-0.063114	9.011300×10^{-2}
-11	-0.041014	3.802500×10^{-2}	-11	-0.041407	2.123200×10^{-1}
-10	-0.038193	1.258500×10^{-2}	-10	-0.069897	5.405400×10^{-2}
-9	-0.038606	2.248900×10^{-2}	-9	-0.120494	3.310000×10^{-3}
-8	-0.037677	3.204400×10^{-2}	-8	-0.100653	1.398000×10^{-2}
-7	-0.050198	3.034800×10^{-2}	-7	-0.069880	4.484900×10^{-2}
-6	-0.062253	1.308010×10^{-1}	-6	-0.067185	4.423100×10^{-2}
-5	-0.039550	1.666800×10^{-2}	-5	-0.099077	6.281000×10^{-3}
-4	-0.037867	9.785000×10^{-3}	-4	-0.109723	1.556000×10^{-3}
-3	-0.050567	1.618000×10^{-3}	-3	-0.137299	1.230000×10^{-4}
-2	-0.073439	1.923000×10^{-3}	-2	-0.137930	2.240000×10^{-4}
-1	-0.110694	7.693000×10^{-3}	-1	-0.175116	2.000000×10^{-6}
0	-0.117002	2.299000×10^{-3}	0	-0.145585	1.140000×10^{-4}

Table 17: Correlations and p -values across lagged days for funding rate as IGM (BTC)

Un-transformed Target			Log-Transformed Target		
Lag (days)	Correlation	p -value	Lag (days)	Correlation	p -value
-20	-0.033038	3.710000×10^{-3}	-20	-0.139269	2.452268×10^{-6}
-19	-0.037257	7.985000×10^{-3}	-19	-0.146008	9.375166×10^{-7}
-18	-0.028218	5.156000×10^{-3}	-18	-0.140242	1.358704×10^{-5}
-17	-0.028585	4.822000×10^{-3}	-17	-0.134762	2.668693×10^{-5}
-16	-0.030807	4.123000×10^{-3}	-16	-0.119937	6.301921×10^{-4}
-15	-0.036034	5.943000×10^{-3}	-15	-0.114389	3.135433×10^{-3}
-14	-0.036014	4.289000×10^{-3}	-14	-0.103671	1.578690×10^{-2}
-13	-0.045171	1.579800×10^{-2}	-13	-0.101732	1.167819×10^{-2}
-12	-0.041761	9.163000×10^{-3}	-12	-0.090109	3.787007×10^{-2}
-11	-0.037728	5.433000×10^{-3}	-11	-0.100005	1.332809×10^{-2}
-10	-0.039430	3.854000×10^{-3}	-10	-0.106779	6.117705×10^{-3}
-9	-0.046967	8.843000×10^{-3}	-9	-0.118402	1.739262×10^{-3}
-8	-0.040760	4.888000×10^{-3}	-8	-0.115449	1.768628×10^{-3}
-7	-0.037877	2.763000×10^{-3}	-7	-0.125667	7.331679×10^{-4}
-6	-0.039623	4.046000×10^{-3}	-6	-0.139024	2.373157×10^{-4}
-5	-0.056609	2.516000×10^{-2}	-5	-0.137546	3.213532×10^{-4}
-4	-0.056544	1.758600×10^{-2}	-4	-0.143106	9.346598×10^{-4}
-3	-0.052421	1.785500×10^{-2}	-3	-0.152114	4.478614×10^{-4}
-2	-0.049941	1.655800×10^{-2}	-2	-0.132472	1.899145×10^{-3}
-1	-0.047825	9.832000×10^{-3}	-1	-0.123571	6.631638×10^{-3}
0	-0.037757	5.495000×10^{-3}	0	-0.112695	4.419365×10^{-3}

Table 18: Correlations and p -values across lagged days for SATZeF as IGM (BTC)

Un-transformed Target				Log-Transformed Target			
Lag (days)	R^2	sign(R)	p -value	Lag (days)	R^2	sign(R)	p -value
-20	-0.000914	--	0.878265	-20	-0.000928	--	0.948910
-19	0.001060	1.0	0.016656	-19	0.001547	1.0	0.086287
-18	0.003355	1.0	0.027506	-18	0.001487	1.0	0.112810
-17	0.002402	1.0	0.037336	-17	0.000004	1.0	0.300148
-16	0.000288	1.0	0.027738	-16	0.002582	1.0	0.093067
-15	-0.000876	--	0.818866	-15	-0.000743	--	0.648098
-14	-0.000922	--	0.933350	-14	-0.000206	--	0.360852
-13	-0.000281	--	0.254554	-13	-0.000925	--	0.996747
-12	-0.000428	--	0.346444	-12	0.000671	1.0	0.201506
-11	-0.000859	--	0.653410	-11	-0.000645	--	0.532361
-10	-0.000684	--	0.673538	-10	-0.000025	--	0.361283
-9	0.001536	1.0	0.084350	-9	-0.000922	--	0.995886
-8	0.000006	-1.0	0.301465	-8	-0.000028	--	0.317229
-7	0.001903	-1.0	0.165956	-7	-0.000457	--	0.444518
-6	-0.000872	--	0.805930	-6	-0.000679	--	0.716143
-5	0.000512	1.0	0.138057	-5	-0.000731	--	0.721239
-4	-0.000229	--	0.307706	-4	-0.000824	--	0.746790
-3	0.000571	-1.0	0.321471	-3	-0.000835	--	0.773654
-2	0.001004	-1.0	0.214937	-2	-0.000264	--	0.408844
-1	0.000157	-1.0	0.360406	-1	-0.000573	--	0.598270
0	0.000296	1.0	0.182394	0	-0.000246	--	0.461187

Table 19: R^2 , sign of correlation, and p -values across lagged days for spot-funding correlation as IGM (BTC)